

Reviewer Pack — Minimal Rank of Configuration Spaces vs Resolution Proof Len...

Entry 09d2de52f642 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

2026-05-23 02:51:45 UTC

Minimal Rank of Configuration Spaces vs Resolution Proof Length for Tseitin Formulas

Entry ID: 09d2de52f642 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-23 02:51:45 UTC

1. Conjecture

Field A (mathematical branch): Configuration Space Theory **Field B** (complexity object): Complexity Theory: Resolution Proof Complexity for Tseitin Formulas

Statement:

[‘For every connected graph G , there exists a configuration space $C(G)$ associated with its geometric realization. The resolution proof length for the Tseitin formula constructed on G is lower-bounded by $2^{\Omega(\text{rank}(C(G)))}$.’, ‘Here, $\text{rank}(C(G))$ denotes the minimal number of parameters needed to define the configuration space $C(G)$.’, ‘This lower bound holds independently of the tree-width of the graph.’]

Rationale (proposer’s reasoning):

[‘Configuration spaces provide a geometric framework that can encode combinatorial problems. By studying the properties of these spaces, we may uncover new insights into the complexity of combinatorial structures like Tseitin formulas.’, ‘The rank of a configuration space has been used in geometric group theory to study the structure of groups and their actions on spaces. If this property could be related to the resolution proof length, it might lead to novel approaches for proving lower bounds on the complexity of computational problems.’]

Taxonomy category: TSEITIN_RES (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 6f4c8431ca6bd0f9

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if, for all generated graphs G with $n \leq 40$ vertices and 30 random seeds, the resolution proof length of the Tseitin formula on G is greater than or equal to $2^{(0.5 * \text{rank}(C(G)))}$ with a mean difference greater than 1 standard deviation from the expected value based on chance.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	1.00	SAFE	SAFE
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def generate_random_graph(n):
    graph = {i: set() for i in range(n)}
    edges = [(i, j) for i in range(n) for j in range(i+1, n)]
    random.shuffle(edges)
    edge_count = 0
    while edge_count < n - 1:
        u, v = edges[edge_count]
        if u not in graph[v] and v not in graph[u]:
            graph[u].add(v)
            graph[v].add(u)
            edge_count += 1
    return graph

def is_connected(graph, n):
    visited = [False] * n
    stack = [0]
    while stack:
        node = stack.pop()
        if not visited[node]:
            visited[node] = True
            for neighbor in graph[node]:
                stack.append(neighbor)
    return all(visited)

def compute_rank(graph, n):
    # Approximate rank using a simple heuristic (number of edges)
    return len(graph) - 1

def tseitin_formula(graph, n):
```

```

clauses = []
for node in range(n):
    if not graph[node]:
        continue
    literals = [f"v{i}" for i in graph[node]]
    clause = ["~" + l for l in literals] + [literals[0]]
    clauses.append(clause)
return clauses

def resolution_length(clauses, n):
    # Simplified resolution algorithm to estimate length
    resolved = set()
    while True:
        new_clauses = []
        for i in range(len(clauses)):
            for j in range(i + 1, len(clauses)):
                clause_i = clauses[i]
                clause_j = clauses[j]
                if any(l in clause_i and "~" + l in clause_j for l in set(clause_i) & set(clause_j)):
                    new_clause = [l for l in clause_i if l not in clause_j] + [l for l in clause_j if l not in clause_i]
                    if len(new_clause) == 1:
                        return len(clauses)
                    new_clauses.append(new_clause)
        clauses.extend(new_clauses)
        if not new_clauses:
            break
    return len(clauses)

def run_trial(seed: int) -> dict:
    random.seed(seed)
    n = random.randint(5, 40)
    graph = generate_random_graph(n)
    if not is_connected(graph, n):
        return {
            "metric_name": "Resolution Proof Length",
            "metric_value": None,
            "instances_tested": 1,
            "conjecture_holds": False,
            "counterexample": "Graph is not connected"
        }
    rank = compute_rank(graph, n)
    clauses = tseitin_formula(graph, n)
    length = resolution_length(clauses, n)
    return {
        "metric_name": "Resolution Proof Length",
        "metric_value": length,
        "instances_tested": 1,
        "conjecture_holds": length >= 2 ** (0.5 * rank),
        "counterexample": ""
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or list(range(2, 37)) # Default to first 30 primes if
    results = []
    for seed in seeds:

```

```

    result = run_trial(seed)
    print(f"TRIAL: {result}")
    results.append(result)

mean_value = sum(r["metric_value"] for r in results if r["metric_value"] is not None) / len(re
std_value = math.sqrt(sum((r["metric_value"] - mean_value) ** 2 for r in results if r["metric_
support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

if all(r["conjecture_holds"] for r in results):
    print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fra
elif support_fraction >= 0.8:
    print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fra
else:
    first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
    counterexample = "Graph is not connected"
    print(f"RESULT: FALSIFIED counterexample=\"{counterexample}\" first_failing_seed={first_fa

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

ces_tested': 1, 'conjecture_holds': False, 'counterexample': 'Graph is not connected'}
TRIAL: {'metric_name': 'Resolution Proof Length', 'metric_value': None, 'instances_tested': 1, 'conje
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--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5d647d35.py", line 116, in <module>
    first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
    ~~~~~^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5d647d35.py", line 116, in <genexpr>
    first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
    ~~~~~^
KeyError: 'seed'

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test crashed before producing any data that could confirm or falsify the conjecture. | next: Investigate and fix the crash in the test code to proceed with the verification of the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	14407
2	preregistration	ollama_remote	glm4:latest	0	0	9570
3	novelty	ollama_remote	glm4:latest	0	0	8348
4	novelty	ollama_remote	glm4:latest	0	0	8645
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	39058
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11737
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8364
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11567
9	judge	ollama_remote	glm4:latest	0	0	12530

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 124228 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/09d2de52f642.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/09d2de52f642.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/09d2de52f642.tar.gz` (if generated)